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The Missing Link in India's Maternal Health Story

SPOTLIGHT

Bindu Shajan Perappadan

Saritha, 27, could not manage to exclusively breastfeed her sons – now aged four and eight – for the recommended first six months despite having normal institutional deliveries. She works as a domestic help in central Delhi's Rajendra Nagar. Her husband drives for a ride-hailing platform and takes up odd jobs such as washing cars and serving at weddings to support the family.

"I returned to work within two months of my delivery and had no option but to introduce infant formula before my sons turned six months old. If I do not work, we cannot pay rent or meet our living expenses," she said.

Saritha is among a growing number of mothers in India unable to exclusively breastfeed (EBF) their infants for the recommended six months.

Why EBF matters

EBF is considered one of the most effective interventions for ensuring the survival and healthy development of the child. Breast milk contains antibodies that protect an infant against common illnesses such as diarrhoea and respiratory infections, supports healthy growth and brain development, and lowers the risk of malnutrition and infant mortality.

The World Health Or-

ganization (WHO) and the United Nations Children's Fund (UNICEF) recommend initiating breastfeeding within one hour of birth and continuing it for the first six months.

Trends in EBF are closely monitored in the National Family Health Survey (NFHS) because they are important indicators of child health, nutrition, and overall public health progress. The latest NFHS-6 indicates worrying trends.

A worrying decline

Despite 90.6% of deliveries in the country now being institutional, fewer infants are exclusively breastfed during the first six months. Although early initiation of breastfeeding has improved from 41.8% to 50.1%, EBF declined from 63.7% in NFHS-5 to 55.8% over the same period, as per NFHS-6 (2023-24).

NFHS-6 data show some of the sharpest declines in exclusive breastfeeding in some of the largest States. In Uttar Pradesh, EBF fell from 59.7% to 34.6% while in Haryana it fell from 69.5% to 41.2%. Delhi witnessed a drop from 64.3% to 54% as well. Rajasthan and Madhya Pradesh also recorded declines. In contrast, Kerala, Gujarat, and West Bengal registered improvements in EBF rates.

What is also worrying is that, as per the survey, the decline in EBF was steeper in rural areas. Data show that EBF among infants below six months fell from 65.1% to 56.2% in rural

The missing link in India's maternal health story

Despite improvements in institutional deliveries and maternity programmes, exclusive breastfeeding rates have declined, according to NFHS-6 data



Healthy start: The World Health Organization and UNICEF recommend exclusive breastfeeding during the first six months to support healthy growth and protect infants from common illnesses. FILE PHOTO

areas. In urban areas, it fell from 59.6% to 54.5%.

A paradox

What makes the decline particularly striking is that it comes at a time when India has made significant gains in maternal and child health.

Early initiation of breastfeeding has improved, maternity benefits have expanded, and women are more financially and digitally empowered than ever before.

Nearly 89% of women now participate in key household decisions while a si-

milar proportion operate their own bank accounts. Digital inclusion has expanded rapidly, with Internet use among women rising from 33.3% in NFHS-5 to 64.3% in NFHS-6. Yet fewer infants are being exclusively breastfed during the first six months.

"The decline in EBF despite these improvements indicates that empowerment alone may not be sufficient. Growing economic pressures, informal employment, lack of maternity protection, inadequate workplace support and limited postnatal assistance

continue to constrain many mothers' ability to exclusively breastfeed their infants for the recommended first six months," said Arun Gupta, paediatrician and public health advocate recognised for his decades-long drive to promote breastfeeding.

Obstacles to EBF

Health experts attribute the decline, particularly in rural India, to increased migration and seasonal labour, early return to agricultural and informal work, changing family structures, erosion of tra-

ditional support networks, greater penetration of packaged foods and formula products, weak postnatal counselling and rising caesarean deliveries.

The sharp rise in C-section rates, from 21.5% to 27.2%, is also cited as a contributing factor. Breastfeeding advocates say surgical deliveries can make early initiation and continuation of breastfeeding challenging, especially in absence of adequate postnatal lactation support.

"Common medical reasons for not being able to exclusively breastfeed include delayed initiation of breastfeeding after delivery, especially following a C-section; perceived or actual low milk supply; lack of breastfeeding counselling and family support; maternal illness or complications; and difficulties faced by premature or low birth weight infants," said Delhi Medical Association member Anil Bansal.

Vandana, 32, became a mother four years ago. An HR consultant at a private firm, she was unable to exclusively breastfeed her child for the recommended six months.

"After my C-section, despite skin-to-skin contact, lactation support and frequent feeding attempts, I could not produce enough milk to meet my baby's nutritional needs. Supplementation became necessary," she said, adding that early counselling may have helped her.

When breastfeeding is

not possible, the WHO recommends infant formula as an alternative. Some hospitals also provide pasteurised donor human milk, particularly for preterm and unwell newborns. However, public health experts stress that formula is intended as a substitute when breastfeeding cannot be sustained, not as a replacement for breast milk.

Policy measures

India promotes breastfeeding through programmes such as Pradhan Mantri Matru Vandana Yojana and Integrated Child Development Services. Comprehensive Lactation Management Centres and Lactation Management Units provide lactation support and donor human milk when the mother's milk is unavailable.

The Infant Milk Substitutes Act, restricts the promotion of infant formula and protects breastfeeding from commercial marketing pressure. However, Dr. Gupta said aggressive marketing of infant formula and substitutes could still undermine breastfeeding and should be strictly regulated. According to Dr. Gupta, breastfeeding is among the most cost-effective public health interventions, even as data expose a gap between policy success and lived reality.

The informal sector

Women employed in the organised sector are entitled to 26 weeks of paid ma-

ternity leave under the Maternity Benefit Act, 1961, amended in 2017. Unfortunately, these protections are largely confined to the organised sector.

According to the e-Shram portal, a national database of unorganised workers, over 16.69 crore women were registered as workers in the unorganised sector as of 2025.

According to health experts, this disconnect raises a fundamental policy question: can India expect mothers to exclusively breastfeed for six months when most receive neither income support nor workplace protection during that period?

Shweta, 25, the mother of a five-year-old girl, said many women working alongside her in a garment factory in Noida, Uttar Pradesh, could not exclusively breastfeed their children. "Rising inflation and the fact that we work on contracts do not allow us the luxury of taking long breaks," she says.

Dr. Gupta says maternity benefits and breastfeeding support systems, particularly for women employed in the informal sector, need to be strengthened. "For the vast majority working in informal occupations, breastfeeding recommendations often collide with economic reality, and these women, who are most in need of support, are often the least likely to receive it," says Dr. Bansal.

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The Missing Link in India's Maternal Health Story

- **The Worrying Trend**

- Despite institutional deliveries rising to 90.6%, exclusive breastfeeding (EBF) rates declined from 63.7% (NFHS-5) to 55.8% (NFHS-6, 2023-24)

- **Why EBF Matters**

- WHO/UNICEF recommend breastfeeding within one hour of birth, continued exclusively for six months; breast milk provides antibodies and lowers risks of malnutrition and infant mortality

- **The Paradox**

- The decline comes despite improved maternity benefits and greater financial/digital empowerment of women, suggesting empowerment alone is not sufficient

- **Structural Obstacles**

- **Rising informal employment, early return to work, inadequate workplace support, erosion of traditional family support systems, and aggressive infant formula marketing all constrain EBF**

- **Policy Gaps**

- Maternity Benefit Act protections remain largely confined to the organised sector; most working women in India's informal sector lack paid leave or lactation support

ISRO Conducts Test of Key Component for Gaganyaan Mission

ISRO conducts test of key component for Gaganyaan mission

The solid motor-based platform SOLVE will be used for parachute tests of the crew module; it has been derived from the PSLV strap-on motor

The Hindu Bureau
BENGALURU

The Indian Space Research Organisation (ISRO) has successfully carried out the first ground test of the solid motor-based Sub-Orbital Launch Vehicle for Experiments (SOLVE) at its Static Test Facility at the Satish Dhawan Space Centre in Sriharikota.

The SOLVE solid motor is a key component for conducting Gaganyaan test missions and the first ground test was carried out on July 3.

The ISRO is developing a solid motor-based SOLVE as a test platform to carry out Integrated Parachute Tests for validation of deceleration system of Gaganyaan's crew module under various test conditions.

During these test missions, the crew module will be carried to an altitude of 10 km to 17 km and separated from the vehicle.

A series of 10 parachutes will be deployed to reduce the velocity of the crew module before splashdown in the sea.



The ground test of the SOLVE solid motor being conducted at the Satish Dhawan Space Centre in Sriharikota. SPECIAL ARRANGEMENT

The solid stage of SOLVE is derived from the Polar Satellite Launch Vehicle (PSLV) Strap-on Motor with a few modifications to meet the Gaganyaan test requirements, including development of slow burn rate propellant and straight nozzle with secondary injection thrust vector control.

The ISRO said that during the test motor performance parameters were as expected. "The development of the SOLVE vehicle provides flexibility in conducting Gaganyaan test missions simulating va-

rious conditions in the actual mission," the space agency said.

Over the past few months, the ISRO has carried out key tests related to the Gaganyaan mission, which include the second Integrated Air Drop Test (IADT-02) and a behavioural study Mission MITRA (Mapping of Interoperable Traits and Response Assessment) in Leh, during which the four Indian astronauts selected for the mission along with scientists, engineers and medical teams were subjected to a week-long study.

ISRO Conducts Test of Key Component for Gaganyaan Mission

- **Gaganyaan is the 1st human space flight programme of ISRO.**
- **Objective:** To demonstrate indigenous capability of human space flight to low earth orbit.
- **Launch vehicle:** GSLV-Mk III, also called the LVM-3 (Launch Vehicle Mark-3).
- **Payloads:**
 - **Crew Module:** It is the spacecraft carrying human beings.
 - **Service Module:** It will support the crew module and is powered by liquid propellant engines.
- **This mission consists of:**
 - **Two unmanned missions (G1 & G2):** 2nd uncrewed mission (G2) will carry “Vyommitra” (a female-looking humanoid robot developed by ISRO to function on-board the Gaganyaan).
 - **One manned mission (G3):** The Orbital Module of the crewed mission will carry three Indian astronauts, including a woman into space for 7 days. It will orbit the Earth at a low-earth-orbit at an altitude of 300-400 km.
- **India will become the 4th nation in the world to launch a Human Spaceflight Mission after USA, Russia, and China.**
- **Russia and France are cooperating with India for Gaganyaan mission.**

India Needs a Second Home for Asiatic Lions

India needs a second home for Asiatic lions

India's conservation of the Asiatic lion is widely celebrated as a remarkable success story. From a population which was reduced to just a few dozen in the early 20th century, the number of Asiatic lions has increased to approx. 891 today. Yet, behind this achievement lies a persistent policy failure: the inability to establish a second, geographically separate population. Scientific institutions, government bodies, and even the Supreme Court have long warned that without such a step, the species remains vulnerable to extinction from a single catastrophic event.

A second dwelling

The need for a second home for Asiatic lions is rooted in decades of government-backed research. Multiple reports of the Wildlife Institute of India have consistently emphasised that a single population confined to one landscape is inherently vulnerable. Studies conducted since the 1980s have concluded that threats such as epidemics, forest fires, or other natural disasters could potentially wipe out the entire species if it remains geographically concentrated.

This scientific consensus was formally acknowledged in the landmark Supreme Court judgment of April 15, 2013, which directed the translocation of Asiatic lions from Gujarat's Gir forest to the Kuno National Park in Madhya Pradesh. The Court made it clear that conservation must be guided by ecological principles rather than regional considerations, stating that a second population was essential for the species' long-term survival. The judgment explicitly recognised the risk that a single calamity could eliminate the entire population if it remained confined to one location.

Despite this clear directive, the translocation project has been stalled for over a decade. Gujarat has consistently resisted the relocation of lions outside the



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Reports by the Wildlife Institute of India have consistently emphasised that a single population confined to one landscape is inherently vulnerable

State, citing its success in conserving the species and raising concerns about habitat suitability elsewhere. This has created a policy impasse between scientific recommendations, judicial mandates, and State-level political considerations. While the Kuno National Park was prepared specifically to host the lions – by relocation of villages and habitat restoration among others – no lions have been introduced there.

Meanwhile, the ecological risks identified by experts have only intensified. The concentration of the entire global population of Asiatic lions within the Gir landscape makes them uniquely vulnerable to disease outbreaks. The 2018 outbreak of the Canine Distemper Virus among the population, which killed several lions and infected many others, serves as a stark reminder of this danger. Such diseases spread more rapidly in dense, socially structured populations, and their impact is amplified by the limited genetic diversity of the lions.

Government-supported studies and conservation plans have repeatedly highlighted that geographically isolated carnivore populations face extinction risks from both biological and environmental factors. In a single-site population, even non-recurring events – such as droughts, wildfires, or sudden prey decline – can have disproportionate effects. This is why conservation science advocates for a "metapopulation

approach", where species are distributed across multiple habitats to reduce risk.

Recognising these concerns, recent policy initiatives such as Project Lion (2020) have attempted to revive the discussion on expanding lion habitats. There have been proposals to develop alternative sites such as the Barda Wildlife Sanctuary within Gujarat itself. However, experts argue that such sites, being geographically close to Gir, may not adequately address the core issue of risk diversification. A second population must be sufficiently distant to prevent the spread of disease or disaster across both populations simultaneously.

The delay in establishing a second home raises broader questions about environmental governance in India. It reflects a tension between national conservation priorities and State-level interests, where wildlife – though constitutionally recognised as a shared responsibility – can become entangled in regional identity and political considerations. The Supreme Court had clearly stated that Asiatic lions are a national heritage, not the property of any single State, yet implementation of this principle remains incomplete.

From success to security

Ultimately, the issue is whether India is willing to move from conservation success to conservation security. The current situation represents a paradox: a thriving population that remains ecologically fragile. Without establishing a second free-ranging population, the gains of decades of conservation effort remain vulnerable to reversal. The case of the Asiatic lion underscores a fundamental principle of modern conservation: numbers alone do not ensure survival; resilience does. Creating a second home is an ecological necessity; continued delay in implementing this vision risks turning a global conservation triumph into a preventable ecological vulnerability.



An Asiatic lioness with her two cubs at the Sakkarbaugh Safari breeding centre in Junagadh in 2025. VIJAY SONEJI

India Needs a Second Home for Asiatic Lions

- **Lion is the second largest cat species in the world.**
- **It is divided into two subspecies: the African lion and the Asiatic lion (Persian or Indian Lion).**
- **African lion (Panthera leo melanochaita) :** Bigger than the Asiatic lion + Males do not live with the females of their pride unless they mate or have a large kill.
- **Asiatic lion (Panthera leo persica) ;** Slightly smaller than the African lion. + Males live with the females of their pride.
- **2025 Lion Census : 891 Asiatic lions**
- **Asiatic lions are confined to the Greater Gir Landscape in Saurashtra, Gujarat, Mitiyala Wildlife Sanctuary, Barda Wildlife Sanctuary**

Questions Surrounding the Quad's Future

Questions surrounding the Quad's future

With the U.S. reversing the U.S. Indo-Pacific Command back to the U.S. Pacific Command, criticisms have been levelled against the relevance of the Quad

DATA POINT

Amit Kumar

Last week, Japanese Prime Minister Sanae Takaichi was in New Delhi for the 16th India-Japan Annual Summit, her first visit to India since assuming office. The two sides reaffirmed their commitment to a free and open Indo-Pacific. The reaffirmation, however, comes at a moment when the idea of the Indo-Pacific appears to be losing its principal champion. On June 16, the Pentagon renamed the U.S. Indo-Pacific Command back to the U.S. Pacific Command, reversing a change it had made in 2018.

The renaming of the Command in 2018, although symbolic, recognised the region's growing significance and acknowledged India's centrality to the U.S.'s regional strategy. The reversal suggests that the region no longer enjoys the priority it once did in Washington. In fact, the National Security Strategy released by the U.S. in November 2025 prioritises the Western Hemisphere over the Indo-Pacific and mentions the Quad only in passing. It also assuages Beijing, which has long resented the formulation. Washington's approach to managing its relations with Beijing is undergoing a change under the current administration.

All of this has put the future of the Quad under the scanner. If Washington retreats from the idea, the platform built around it inevitably invites questions about its relevance. These anxieties only add to a set of older concerns. How relevant or securitised is the Quad? Is it doing enough? And the criticisms aren't without substance.

The first criticism entails the hesitation to name China. If the objective is indeed to shape the balance of power in the region, the silence over Beijing's military and economic aggression is bad optics. However, this is largely a normative criticism. One can argue that

the Quad's reluctance to name Beijing is strategic, because positioning itself as an anti-China platform could dissuade smaller countries from associating with it.

Critics highlight the Quad's hesitation to project itself as a security-centric grouping as another of its shortcomings. Given that Beijing poses hard security challenges to the region, any balancing effort has to be primarily security-centric. The extent of the grouping's hesitation is evident in the members' active effort to make an implicit distinction between Quad and the Quadrilateral Security Dialogue (QSD). There is a reluctance to use them interchangeably, even as the members acknowledge the grouping's origins to be rooted in the first QSD in 2004.

The Quad's reluctance to elevate the partnership to a 2+2 format – a bilateral ministerial dialogue comprising Foreign and Defence Ministers – is another pain point. Even as each of the four countries has formalised 2+2 meetings bilaterally with one another, the Quad continues to host meetings only at the leaders' and Foreign Ministers' level.

Domain of cooperation

The alleged overwhelming focus on non-security or soft issues is a related criticism. However, a review of the Quad's agenda tells a different story. Even though the Quad assumed its current form in 2017, there was a lull for the first four years. It only picked up steam in 2021 during the COVID-19 pandemic, when the four countries held their first virtual leaders' summit and announced a first set of initiatives. Within six months, the Quad held its first in-person leaders' summit and expanded its areas of cooperation. The list today has grown to 37, of which 23 fall under security-related domains (Chart 1). Technology sovereignty, maritime capabilities and secure communications will be key in countering China in the years ahead.

Choosing the right areas of cooperation doesn't necessarily reveal much, though. The type of cooperation and capital flows are key determinants in evaluating the seriousness of the grouping. But a review of the 'type' of cooperation in security-related domains suggests that Quad has a heavy focus on interoperability, data sharing, monitoring, and relief (Chart 2). While this is necessary and critically important for any security-focused grouping, Quad's integration efforts are limited to civilian or non-military grade applications. Nonetheless, these integration exercises have dual-use capabilities and can be leveraged for even military-grade purposes.

The Quad's focus on building capacity and resilient supply chains is also significant, especially as these relate to critical and emerging technologies, maritime, and communications – the domains where the bulk of contestation with China is expected.

The Quad's security focus is not matched by its funding pattern. Based on what is available in the public domain, health and climate have been the primary beneficiaries (Chart 3). One can argue that since most security-related initiatives focus on integration and standard-setting, they do not require the same scale of dedicated funding. But building resilient supply chains would require substantial capital commitment, and that is where the gap is most visible.

The Quad's recent commitment to mobilise \$20 billion towards the Critical Minerals Initiative and Framework could change this picture. But it remains only a stated target at this point, with no funding committed or allocated. If security issues make up close to two-thirds of Quad's initiatives, the funding allocation must follow suit.

The questions surrounding the Quad's future will not be settled by nomenclature in Washington. They will be settled by whether the grouping backs its security-heavy agenda with capital.

Uncertain course

Amit Kumar is a Staff Research Analyst with the Geostrategy Programme at The Takshashila Institution. The data for the charts were sourced by the author from his Quad Monitor dashboard



Looking forward: External Affairs Minister S. Jaishankar with U.S. Secretary of State Marco Rubio, Australian Foreign Minister Penny Wong and Japanese Foreign Minister Toshimitsu Motegi during the Quad Foreign Ministers' Meeting, in New Delhi on May 26, 2025. ANI

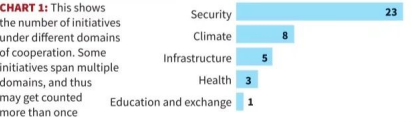


CHART 3: This shows the funding for initiatives under different domains of cooperation. Health receives the highest funding, followed by climate at a distance second

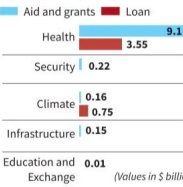
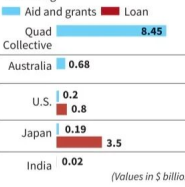


CHART 4: This shows the funding for various initiatives of the Quad by different countries and the Quad Collective. Absence of values indicate no funding



Questions Surrounding the Quad's Future

- **The Trigger**

- On June 16, the Pentagon renamed the U.S. Indo-Pacific Command back to the U.S. Pacific Command, reversing a 2018 change and raising questions about Washington's continued prioritisation of the Indo-Pacific

- **Symbolic Shift**

- The U.S.'s November 2025 National Security Strategy prioritises the Western Hemisphere over the Indo-Pacific and mentions the Quad only in passing

- **Criticism : Reluctance to Name China**

- Despite Beijing posing the region's primary security challenge, the Quad avoids explicitly naming China and hesitates to fully elevate itself as a security-centric grouping

- **Domain of Cooperation**

- **Of 37 total initiatives, 23 fall under security-related domains, focused on interoperability, capacity building and supply chain resilience**

- **Funding Mismatch**

- Health and climate have received the most funding despite security comprising nearly two-thirds of initiatives; the \$20 billion Critical Minerals Initiative remains only a stated target with no committed funding yet

E-rickshaw Battery 'Hacks' and Bluetooth BMS Vulnerabilities



Almost all of the components that make up e-rickshaws are imported from China, [see](#)

E-rickshaw battery 'hacks' and Bluetooth BMS vulnerabilities

The misuse of battery management apps exposed security flaws in imported e-rickshaw battery systems, rather than problems with the apps themselves stronger supply chain oversight and security standards, not app bans, are needed to address such vulnerabilities

Karan Saini

The story so far: Over the last week, social media has been flooded with videos depicting individuals using mobile applications such as **BAT BMS**, **Lossy** and **Epch-Ion** to wirelessly disable the batteries of e-rickshaws using Bluetooth. The videos have largely been shared as "prank" videos, however cruel, unethical and unlawful they may be – while some have speculated that the applications were designed for high-tech sabotage by China's manufacturers. The reality is far less dramatic, yet still worth understanding carefully.

How are battery management systems diagnosed? Applications such as BAT-BMS exist to allow fleet owners and service technicians to easily diagnose battery problems such as overvoltage or undervoltage. In most vehicles, battery management and diagnostics require physical access to the module and the use of a computer or other device to gather information. Over time, manufacturers have begun equipping these modules with Bluetooth capabilities, making diagnostics and repairs easier without requiring extensive disassembly. Applications such as BAT-BMS are typically developed alongside the battery management systems they support and are almost certainly not intended to sow chaos by disabling as many vehicles as possible. Instead, they search for specific devices advertising themselves as battery management modules over Bluetooth, connect to one device at a time, display

diagnostic information, and provide basic debugging functions.

How are e-rickshaws built and why imported electronics matter? The BAT-BMS application is specifically designed to discover and connect to **Battery Management System (BMS)** modules manufactured by a Chinese electronics company, **Jiabaida**. A seven-year-old public discussion on the open source code sharing platform **GitHub** documented a lack of authentication in **Jiabaida BMS** modules with the author even providing a hacky workaround for this obvious security flaw. Over time, this vulnerability appears to have been discovered and subsequently exploited by individuals who began popularising its misuse. In particular, some users circulated short-form "prank" videos demonstrating how it could be used to disable e-rickshaw batteries.

Almost all of the components that make up e-rickshaws are imported from China. In fact, one could order almost all the parts required to assemble one from consumer websites such as **Alibaba Group** or **AliExpress**. Of course, one could go even further and import a fully pre-assembled e-rickshaw from **China** through these platforms, though the shipping fees and Customs duties might eat into your margin. Most manufacturers import the electronic components required to assemble e-rickshaws from **China**, leaving only the fabrication and assembly of the aluminium frame and chassis to be done domestically.

When technology is blamed for system vulnerabilities It appears that whenever something goes

wrong, the government is quick enough to attribute blame to digital platforms. Telegram was ordered to be blocked during the re-NET period, reportedly to protect individuals from fraudsters claiming to sell leaked examination question papers in advance. Similarly, following viral "prank" incidents, the government directed Google and Apple to remove several popular battery management apps from their respective app stores.

In both cases, the government has chosen to restrict access to applications rather than address the underlying issues in a meaningful way. While the two applications are fundamentally different – one being a widely used communication platform with social media features, and the other a set of utilities for diagnosing battery issues – the reasoning behind both decisions appears similar.

In essence, when a particular technology becomes associated with harm, restricting access to that technology is seen as a way to eliminate the harm itself.

Hypothetically, if the re-NET examination were to be compromised, the breach would have to occur at the source or during transit – during question formulation, printing, or transport. The breach would not originate from, nor be enabled by, the Telegram platform itself. Blocking Telegram in this hypothetical scenario would only temporarily disrupt the sale of leaked question papers until sellers migrate to another platform of choice. Similarly, restricting battery management applications does not make vulnerable BMS modules any more secure. If anything, it might further inconvenience users who have been

affected and are trying to restore functionality to their vehicles.

How should supply chain regulation be structured?

Contrary to popular belief, regulatory frameworks governing e-rickshaws do exist. Gazette notifications **GSR 70903** dated 8.10.2014 and **S.O. 259007** dated 8.10.2014 define requirements related to overall dimensions, tyres, brakes, horns, and vehicle identification number standards, but do not provide detailed standards for electronic components.

Globally, as security risks become increasingly associated with the supply chains, various strategies have been adopted to address vulnerabilities before a product reaches the assembly line. These include vetting suppliers, auditing the firmware of electronic components, and tracing the origin and capabilities of individual parts. Such measures can be mandated through regulation. A simple Google search conducted seven years ago could have alerted e-rickshaw manufacturers to the security issues in the BMS modules they use.

Frameworks that establish standards, as well as responsibilities and liabilities for manufacturers, should ideally extend to all products containing electronic components, not just vehicles or specifically e-rickshaws. While sourcing parts from the cheapest available source might be suitable for building a quick prototype, the long-term production and marketing of products should require greater effort and due diligence than simply placing a few orders and setting up an assembly line.

(Karan is an independent security researcher from New Delhi, India)

E-rickshaw Battery 'Hacks' and Bluetooth BMS Vulnerabilities

- **The Trend**

- Social media videos have shown individuals using apps like BAT-BMS to wirelessly disable e-rickshaw batteries via Bluetooth, largely shared as "pranks"

- **The Real Cause**

- The vulnerability lies in Battery Management System (BMS) modules made by a Chinese electronics company, Jiabaida, which lacked proper authentication — a flaw publicly documented on GitHub seven years ago

- **Import Dependency**

- Almost all e-rickshaw electronic components, including BMS modules, are imported from China; only frame fabrication and assembly happen domestically

- **The Governance Response**

- Government directed Google and Apple to remove several battery management apps from app stores, restricting access to technology rather than addressing the underlying supply-chain vulnerability

- **The Real Fix**

- Existing 2014 gazette notifications regulate e-rickshaw dimensions, tyres, brakes and VIN standards but not electronic component security; experts call for supply chain oversight rather than app bans

Can Bar Associations Refuse to Represent an Accused?

Can Bar Associations refuse to represent an accused?

What has the Supreme Court said about Bar resolutions denying legal representation?

LETTER & SPIRIT

Rizmi Lia M.

The story so far:

The Faizabad Bar Association's resolution to not represent the accused in the Ayodhya Ram Temple embezzlement case has prompted the question of whether a Bar Association could collectively decide not to represent a particular accused.

What is the Supreme Court's take?

The Supreme Court has consistently held that every accused has the right to fair trial and legal representation and it is illegal, unconstitutional and contrary to professional ethics to deny that right.

In a Supreme Court judgment in 2010, *A.S. Mohammed Rafi v. State of Tamil Nadu*, the case arose out of a confrontation between lawyers and police personnel in Coimbatore.

Following the incident, a local Bar Association passed a resolution that none of its members would represent the accused police personnel. A Division Bench of Justices Markandey Katju and Gyan Sudha Misra held that such resolutions by Bar bodies were "wholly illegal, against all traditions and professional ethics".

The court declared in the judgment that "all such resolutions of Bar Associations in India are null and void and the right-minded lawyers should ignore and defy such resolutions if they want democracy and rule of law to be upheld in this country".

What is the accused's right to be defended?

Article 22(1) of the Constitution guarantees that no arrested person shall be denied the right "to consult, and to be defended by, a legal practitioner of his choice". Article 14 provides for equality before the law and equal protection of the

laws within the territory of India.

The Supreme Court has recognised the right to a fair trial as an integral part of the right to life and personal liberty under Article 21.

Article 39A, a Directive Principle of State Policy, further requires the State to ensure that the legal system promotes justice on the basis of equal opportunity and that no citizen is denied access to justice because of economic or other disabilities, including through free legal aid.

What do the Bar Council of India Rules say?

The 'Standards of Professional Conduct and Etiquette' chapter of the Bar Council of India Rules says that "an advocate is bound to accept any brief in the Courts or Tribunals or before any other authorities in or before which he proposes to practise at a fee consistent with his standing at the Bar and the nature of the case. Special circumstances may justify his refusal to

accept a particular brief."

However, the Uttarakhand High Court in *Kuldeep Agarwal v. State Of Uttarakhand And Others* (2019) clarified that "special circumstances mentioned in Clause II, justifying refusal of an advocate to accept a particular brief, refers, by the use of the word "his", to the advocate in his individual capacity, and not to the Bar Association whose members are advocates".

Have such resolutions been passed before?

Yes. Following the 2008 Mumbai terror attacks, a Bar resolution opposed legal representation for arrested terrorist, Ajmal Kasab. The legal aid lawyer initially assigned to him declined to appear, while another advocate who agreed to defend him faced political threats before a lawyer was eventually appointed under police protection.

After the December 16, 2012 Delhi gangrape case, lawyers at the Saket courts passed a similar resolution refusing to represent the accused. Likewise, lawyers declined to defend the accused in the 2019 Hyderabad veterinary doctor's rape and murder case.

How have courts dealt with such resolutions?

The Uttarakhand High Court had in 2019 declared null and void a resolution passed by the Kotdwar Bar Association

threatening to terminate the membership of any lawyer who represented an accused in the murder of an advocate.

In 2020, the Karnataka High Court criticised the Hubballi Bar Association after lawyers objected to advocates representing Kashmiri students accused of raising pro-Pakistan slogans. The Bench described the conduct of lawyers on the Dharwad court premises who prevented advocates from filing bail applications as "sheer militancy". It also indicated that such resolutions could amount to criminal contempt.

More recently, the Madras High Court in *Manikandan Nair v. State of Tamil Nadu* in 2025 reiterated that Bar Associations cannot formally or informally prevent advocates from appearing for any accused.

In the case of *J. Jayalalitha v. State of Karnataka* (2014), the Supreme Court observed that "fair trial is the main object of criminal procedure and such fairness should not be hampered or threatened in any manner. Fair trial entails the interests of the accused, the victim and the society. Thus, fair trial must be accorded to every accused in the spirit of the right to life and personal liberty and the accused must get a free and fair, just and reasonable trial on the charge imputed in a criminal case".

Thus, the Supreme Court has consistently held that the right to legal representation is a core feature of a fair trial.

Can Bar Associations Refuse to Represent an Accused?

- **The Trigger**

- The Faizabad Bar Association's resolution not to represent the accused in the Ayodhya Ram Temple embezzlement case revived the question of whether Bar bodies can collectively deny legal representation

- **Constitutional and Legal Basis**

- Article 22(1) guarantees the right to consult and be defended by a legal practitioner of choice; Article 21 (right to fair trial) and Article 39A (free legal aid) reinforce access to justice

- **Supreme Court's Consistent Position**

- In A.S. Mohammed Rafi v. State of Tamil Nadu (2010), the Court held that Bar resolutions denying representation are "wholly illegal, against all traditions and professional ethics" and are null and void

- **Bar Council of India Rules**

- Advocates are professionally bound to accept any brief at a fee consistent with their standing; only limited "special circumstances" in an individual capacity justify refusal, not collective Bar body resolutions

- **Recurring Pattern**

- Similar resolutions followed the 2008 Mumbai attacks, the 2012 Delhi gangrape case and the 2019 Hyderabad case; courts (Uttarakhand HC 2019, Madras HC 2025) have repeatedly struck down such resolutions

AI Governance and a Voice for the Global South

AI governance and a voice for the Global South

In February 2026, India hosted the India AI Impact Summit 2026, that sought to put the needs and challenges of the Global South at the centre of the Artificial Intelligence (AI) discourse. The summit's themes framed by India were rooted in the contextual realities of the Global South with a focus on real-world harms. This was a departure from the previous summits (at Bletchley Park, 2023, U.K.; Seoul 2024, and Paris 2025) that prioritised both catastrophic and existential risks over questions of present harms, equity and inclusion.

As the Summit evolved, the political and policy momentum shifted toward raising capital for AI development in India and accelerating adoption through domestic use cases. In this process, India increasingly began to position itself within the newly framed "middle power" discourse, at the cost of Global South solidarity which underpinned the Summit's original vision. India clarified this stance by joining Pax Silica, signalling strategic alignment with the United States-dominated semiconductor supply chain. As part of the agreement, India agreed to adopt a pro-innovation regulatory approach, thereby compromising its pursuit of strategic autonomy.

Middle power dilemma

This repositioning of India's geopolitical character, as a middle power, has left it in a lonely corner. The middle power narrative is diplomatically attractive but strategically uneasy. India's aspirations to be positioned alongside European and Asian countries such as Japan, which do not consider India a peer in technological capability or economic development, is also in dissonance with its colonial past and low per capita – realities that firmly anchor India within the Global South.

Accompanying the friction between India's ambition and its realities is the U.S.'s foreign policy push for global AI adoption of U.S. tech, bringing into question whether it will be a reliable



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India faces a
choice between
dependency and
leadership in AI
governance

partner in the AI adventure to India. The U.S. has declared its disinterest in AI governance, especially global multilateral or multistakeholder governance. This raises fundamental questions for India and the Global South on concentration of infrastructure, and economic power in the U.S. Will this be a repeat of the social media story – when U.S. foreign policy pushed back against regulating for user harms to safeguard the interests of social media platforms concentrated in the U.S.? Furthermore, economic value primarily accrues to American industry despite significant business and users outside its borders, while disproportionate externalities and harms also persist in domestic markets.

Pertinent issues

In the AI story, numerous questions arise. Will India mainly be a consumer of U.S. tech with Indian users bearing disproportionate harm? Will India be a site for extraction of data, labour for data labelling, minerals for manufacturing, and land, water, electricity and resources for data centres, primarily enabling the growth of American Big Tech?

Since the summit in February, India has sanctioned land for data centres displacing communities, triggering protests. There are no meaningful guardrails to protect local communities as American companies scrape public content to build language and indigenous knowledge datasets. The non-profit ecosystem is signing memoranda of understanding to diffuse AI and adopt use cases. However, fundamental AI innovation has been slow – India remains unable to compete with global foundational models, its semiconductor development is focused on low-value assembly and there is a question of adequate capital to invest and grow the national AI ecosystem.

But perhaps all is not lost for India. The first of a two-part UN [Global Dialogue on AI](#) is underway in Geneva (July 6-7, 2026). Stakeholders will

convene to discuss how the multilateral and multistakeholder ecosystem can come together to collectively define the rules for the governance of AI.

A window for leadership

India can use the opportunity to stitch together a fractured AI policy agenda that currently lacks a leader. It remains one of the few countries with the political heft, the technical capacity, and a diverse market to play this role. Rather than positioning itself merely as a destination for investment or a market for AI, India could reassert a vision of technological development rooted in public purpose, user safety, strategic autonomy, and international cooperation.

India should reiterate the need for international norms that empower Global South countries to focus on building local AI ecosystems and fostering innovation, safeguarding users, enhancing regulatory capacity, enabling skilling and developing domestic infrastructure. It should also advance critical debates on competition and consumer protection and ensure economic value accrues within national markets.

Concurrently, India must create pathways for international cooperation on AI within the Global South. The Geneva dialogue is a critical moment for Global South countries to come together to enable enhanced agency and strategic autonomy.

This requires developing innovative approaches to pool capacity and resources including cooperation on data, compute, interoperable standards and shared protocols and governance, and strengthening institutional capacity both regulatory and technical across the Global South.

As heterogeneous as the Global South is, it can be a counterweight to the hegemony of Big Tech. India can lead this march to ensure that shared governance norms are created and benefits are shared with the people in the Global South, appropriately protecting them from harm.

AI Governance and a Voice for the Global South

- **The India AI Impact Summit 2026**

- Held in February 2026, it sought to centre Global South needs and real-world harms in AI discourse, departing from earlier summits (Bletchley Park 2023, Seoul 2024, Paris 2025) that prioritised catastrophic and existential risks

- **The 'Middle Power' Dilemma**

- India's post-summit shift toward capital-raising and domestic AI adoption, alongside joining the US-aligned Pax Silica semiconductor initiative, has been seen as diluting its original Global South solidarity positioning

- **Key Concerns Raised**

- Whether India risks becoming primarily a site for data, labour and resource extraction benefiting foreign Big Tech, echoing patterns seen in the earlier social media era

- **Ground-Level Tensions**

- Since February, land sanctioned for data centres has displaced communities and triggered protests, with limited safeguards; India's foundational AI and semiconductor innovation remains focused on low-value assembly

- **A Window for Leadership**

- The UN Global Dialogue on AI (Geneva, July 6-7, 2026) offers India an opportunity to champion Global South-oriented AI governance norms on data, compute, standards and shared protocols

Thank You